

INSTITUT D'ENSEIGNEMENT SUPÉRIEUR DE RUHENGARI (INES)

SWE 3513 Artificial Intelligence – Assignment 1

Musanze Harvest Link Cooperative

Staff Decision Dashboard: UI/UX Specification

Group Code: AI-G09

Dataset: AI_A1_G09.csv (SHA-256: 1de2293c3b42904d...) Document: AI_A1_G09_UIUX.pdf

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Stakeholder: Musanze Harvest Link Operations Team & INES Assessors

Group Members:

NSHIMIYIMANA Saidi (25/27573) - Data & UX Lead (Group Leader)

KANYANGE Kellen (25/27821) - Regression Engineer

MUGABO Alvin Marvin (24/26657) - Classification Engineer

HAMID ABAASHAMID (25/27817) - Clustering & QA Engineer

TUYISINGIZE Devotha (25/27747) - Reproducibility & Release Lead

1. OPERATIONAL CONTEXT & THREE CORE DECISIONS

Musanze Harvest Link Cooperative coordinates Irish potato collection across volcanic hillside farms in Rwanda (Kinigi, Nyange, Cyanika). Before dispatch to Kigali wholesale markets, staff make three time-critical decisions:

Decision 1 (Regression): Estimate expected harvest weight (actual_yield_kg) to allocate appropriate freight tonnage.

Decision 2 (Classification): Flag consignments needing dispatch attention (0 or 1) to isolate high-risk batches before loading.

Decision 3 (Clustering): Group collection points into operating profiles (k=2..5) to optimize truck routing without claiming natural farm categories.

USER JOURNEY & INFORMATION FLOW

STAGE 1: Field Harvest Ingestion

Extension agents record farm metrics (plot_area_ha, rainfall_mm, soil_ph, seed_kg, distance_km, arrival_hour).

Pipeline calculates SHA-256 fingerprint, verifies zero missing values, and excludes record_id.

STAGE 2: Multi-Model Automated Inference

NumPy Linear Regression computes expected yield (kg) with prediction intervals

Interpretable Logistic Classifier computes dispatch attention probability and binary flag. Unsupervised K-Means assigns consignment to an operational logistics profile.

STAGE 3: Dispatch Supervisor Dashboard Review

Supervisor views triage summary: safe consignments glow green (Standard Dispatch); flagged lots glow amber with actionable root causes (e.g. late arrival + rain).

STAGE 4: Informed Human Action & Override

Supervisor directs flagged lots to Bay B for a rapid 5-minute moisture/temperature audit. Dispatches cleared bulk trucks along clustered routes to Kigali wholesale depots.

ANNOTATED WIREFRAME: INGESTION & YIELD ESTIMATION Annotation A: Data Quality & Fingerprint Header]

Group: AI-G09 Displays SHA-256 fingerprint, row count (420), and schema validation badge.

[Annotation B: Expected Yield Prediction Card]

Shows predicted harvest weight: 18,450 kg $\pm$ 510 kg.

NumPy First-Principles gradient descent model ($R^2 = 0.923$, RMSE = 5,159 kg).

[Annotation C: Truck Fleet Allocation Recommendation]

ANNOTATED WIREFRAME: DISPATCH WARNINGS & CLUSTERS

[Annotation D: Dispatch Attention Alert Banner]

Color-coded high-contrast card: 'FLAGGED FOR PRIORITY INSPECTION'. Probability: 78.4%

Root drivers: Arrival at 17:00 + High rainfall.

[Annotation E: Cluster Operating Profile Card]

Assigns consignment to Profile Cluster 1 (Commercial hillside farms).

[Annotation F: Human Action Buttons]

'Route to Inspection Bay B' vs 'Manual Supervisor Override'.

RESPONSIBLE AI STATES & HUMAN OVERRIDE

1. UNCERTAINTY COMMUNICATION:

Yield estimates include statistical error bounds ($\pm$ RMSE) so dispatchers do not over-commit warehouse capacity.

2. MISSING & MALFORMED SENSOR HANDLING:

predict.py strictly rejects incomplete records with clear JSON diagnostics (e.g., `MissingFieldsError: ['soil_ph']`) to prevent database corruption.

3. MODEL BOUNDARIES & OUT-OF-DISTRIBUTION WARNINGS:

Models are explicitly scoped to volcanic soil in Musanze. Predictions for non-regional varieties trigger an out-of-distribution advisory.

4. HUMAN OVERRIDE GOVERNANCE:

The AI decision is assistive; human supervisors retain final authority with auditable override logging.

VISUAL SYSTEM RATIONALE & THREE DECISIONS

ACCESSIBLE VISUAL SYSTEM (WCAG AA COMPLIANT):

High-contrast emerald (#10b981) for standard clearance, amber (#f59e0b) for attention. Dual encoding: Text badges always accompany color cues for colorblind operators.

THREE METHODOLOGICALLY JUSTIFIED DESIGN DECISIONS:

1. Prominent Error-Cost Visualization: Explains that False Negatives cost 250,000 RWF in transit spoilage vs False Positives costing only 3,000 RWF in quick visual audits.
2. Scientific Clustering Safeguard: Explicitly badges clusters as 'Exploratory Profiles' rather than ground-truth categories.
3. One-Click Verification Terminal: Enables academic assessors to verify all 10 tests reproducibly from a clean command prompt.